First order Logic Exercises 1

1. Mary loves everyone. [assuming D contains only humans]
2. Mary loves everyone. [assuming D contains both humans and non-humans]
3. No one talks. [assume D contains only humans unless specified otherwise.]
4. Everyone loves himself.
5. Everyone loves everyone
6. Everyone loves everyone except himself
7. Every student smiles.
8. Every student except George smiles.
9. Everyone walks or talks.
10. Every student walks or talks.
11. Every student who walks talks
12. Every student who loves Mary is happy
13. Every boy who loves Mary hates every boy who Mary loves
14. Every boy who loves Mary hates every other boy who Mary loves.

First order Logic Exercises 2

1. Every child loves Santa.
2. Everyone who loves Santa loves any reindeer.
3. Rudolph is a reindeer, and Rudolph has a red nose.
4. Anything which has a red nose is weird or is a clown.
5. No reindeer is a clown.
6. Scrooge does not love anything which is weird.
7. (Conclusion) Scrooge is not a child.

First order Logic Exercises 3

1. Every child loves anyone who gives the child any present.
2. Every child will be given some present by Santa if Santa can travel on Christmas eve.
3. It is foggy on Christmas eve.
4. Anytime it is foggy, anyone can travel if he has some source of light.
5. Any reindeer with a red nose is a source of light.
6. (Conclusion) If Santa has some reindeer with a red nose, then every child loves Santa.